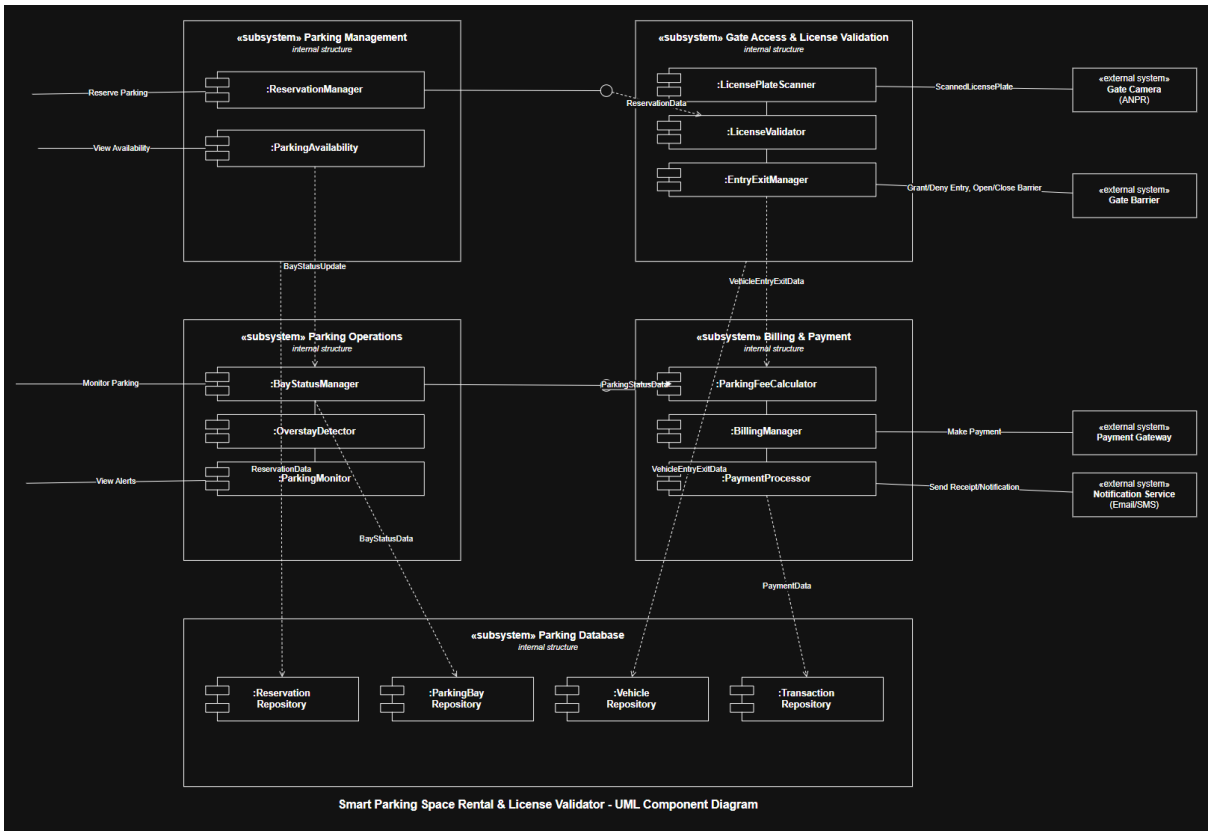
Software Engineering Lab - 3

Lab 3: Component Modelling & Architectural Pattern Selection

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Project Title: Smart Parking Space Rental & License Validator



Design Selection: The architecture is broken down into layers like Parking Management, Parking Operations, Gate Access and License Validation, Payments and Billing and Parking Database.

Point 1: Layered architecture separates the different tasks such as making the reservation, verifying the vehicle entry, managing payments and helping in operating the database which makes it easier to develop and maintain.

Point 2: The layered architecture also allows easy integration of many external systems like the ANPR Gate Camera, Gate Barrier, Payment Gateway and Notification Service without the need to tie it closely with the database.

Security Benefit: The use of layers makes it possible to control access to sensitive tasks like license validation, money receipt management and database transactions and limit their access.

Performance Benefit: Since all tasks are executed separately by the modules, like checking the space available for parking and making payments, the system works faster and does not waste time on unnecessary parts of the operations.